BADR ALDEEN AL-KHAZAN

Operations Research & Supply Chain

Yogyakarta, Indonesia | badr.alkhazan@gmail.com | My Portfolio

SUMMARY

Final-year Industrial Engineering student (GPA 3.90/4.00), Future Global Leaders Scholarship awardee and CAPM-certified, with a research focus on Operations Research and supply chain analytics. Research on supply chain simulation, risk-aware optimization, vehicle routing, humanitarian logistics, and predictive maintenance has been accepted in a journal and at four international conferences. Fluent with AI tools to accelerate modelling, analysis and literature review.

Research interests: stochastic modelling and simulation, vehicle routing, supply chain risk and resilience, decision analytics.

EDUCATION

B.Eng. Industrial Engineering (International Program), Universitas Islam Indonesia *Expected 2027*

Scholarship: Awardee of the Future Global Leaders Scholarship (FGLS), 2023–2027.

GPA: 3.90/4.00. **Relevant coursework:** Operations Research I & II, System Simulation, System Modeling, Supply Chain System, SCM Performance Management, Data Analytics, Statistics I & II, Production Planning and Control, Research Methodology.

RESEARCH & PUBLICATIONS

Where Floods Strike and How Roads Fail: Testing Two Assumptions of Relief Pre-Positioning on the 2025 Cyclone Senyar Floods in Aceh
16th AIC 2026 (Universitas Syiah Kuala), accepted for presentation

Simulation-Based Analysis of the Bullwhip Effect in a Multi-Echelon Bottled Water Supply Chain
JTMIT, accepted, in production, 2026

When Does Risk-Aware Workload Balancing Pay Off? A Three-City Last-Mile Routing Benchmark Using Industrial Traces
ICORSI 2026 (IIT Delhi, 21–23 Dec 2026), abstract accepted

Safe Capacity and Time Compliance in School-Meal Kitchens: Evidence from Indonesia's Free Nutritious Meal Program
Spektrum Industri, under review, 2026

A Configurable Electric Vehicle Routing Framework for Urban Retail Delivery with Public Charging: A Yogyakarta Case Study
In progress, 2026

Balancing Cost and Tail Risk in the Indonesian Wheat Import Portfolio Using Nested Mean CVaR Optimization
ICOBLAST 2026 (Diponegoro University), full paper accepted

Supply Chain Governance in Sustainability Reporting: a SCOR Orchestrate Analysis of an Indonesian State-Owned Enterprise
IJPMB, under review, 2026

Calibration and Cost-Sensitive Threshold Selection for Imbalanced Predictive Maintenance: an AI4I 2020 Study
TMIC 2026 (4th Trunodjoyo Madura International Conference), presented, Best Presenter Award

Reducing University Motorcycle Exit Time through Automated Parking Access Control (SIGAP)
JuTISI, under review, 2026

How Depending on AI Can Impact the Analytical Skills and Critical Thinking of Young People
Published, 2026

Germany's DIGITAL Cooperative Framework: Advancing Sustainable, Inclusive, and Innovative Global Digital Infrastructure
Published, 2025

AWARDS & CERTIFICATIONS

- **Future Global Leaders Scholarship**: Universitas Islam Indonesia (2023–2027)
- **Outstanding Delegate**: IYEC 12, Malaysia & Singapore
- **Best Presenter**: TMIC 2026
- **Best Presentation**: EXPO RSKE 2025, Multifunctional Ergonomic Pen
- **TOEFL iBT 90/120**: ETS (2026–2028)
- **CAPM®**: Project Management Institute (2026–2029)
- **1st Place**: EXPO PSIT 2026, SIGAP RFID Parking Access
- **1st Place**: Idea Champion 2.0, AI-PHL Guardian

EXPERIENCE & LEADERSHIP

Simulation & Modelling Trainee, Delsim Laboratory, Universitas Islam Indonesia *Apr 2025 – Sep 2025*

- Built and validated discrete-event and system-dynamics models using FlexSim and PowerSim to analyze industrial system behavior.

Industrial Company Visit, PT Gaya Motor (BMW Indonesia), organized by Universitas Islam Indonesia *Apr 2025*

- Toured the assembly line and studied quality assurance systems, from parts inspection to final testing.

Deputy Head, Events Division & Industrial Engineering Ambassador, UII Global (Volunteer) *Jan 2025 – Dec 2025*

- Led academic and cultural events, logistics and partnerships; served as PIC for international-student activities.

Deputy Head, Social Responsibility Circle, Yemeni Students' Union in Indonesia (Volunteer) *Apr 2025 – Mar 2026*

- Coordinated student-support and community programs and major cultural events.

SKILLS & LANGUAGES

Methods: stochastic and mean-CVaR optimization, vehicle routing, queueing analysis, discrete-event simulation, system dynamics.
Tools: Python (OR-Tools, SciPy, pandas), FlexSim, PowerSim, Advanced Excel, generative AI tools.
Professional: Project management, team leadership, cross-cultural communication, public speaking, technical writing.
Languages: Arabic (Native), English (B2), Indonesian (B2), German (A1).